



EMERITUS

Learn. From the world's best.

Convolutional Neural Networks - I

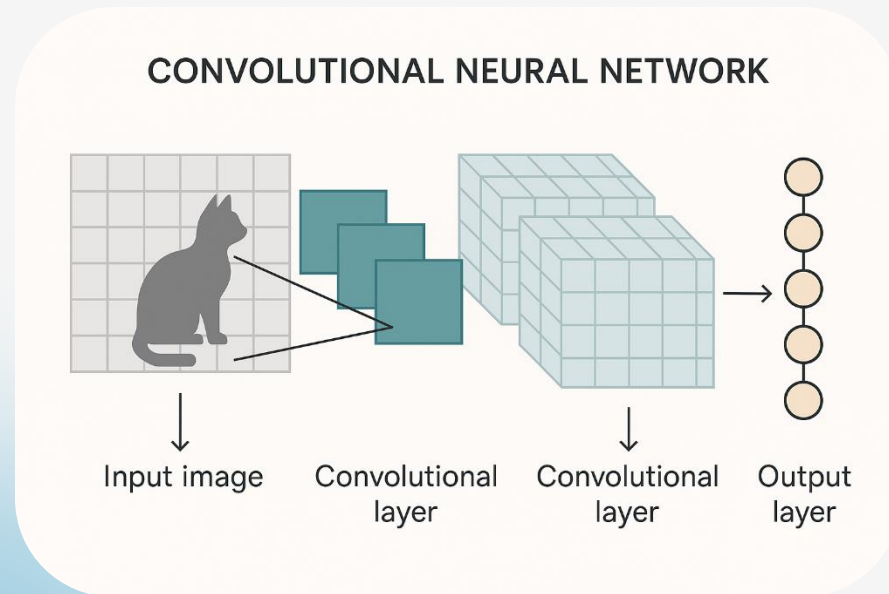


EMERITUS

Learn. From the world's best.

Convolutional Neural Networks

Neural Networks For Visual Data



Inspired by the human visual system

Designed to handle spatial data efficiently

Automatically learn hierarchical features from images

Transform how machines interpret visuals

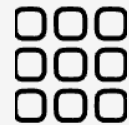
Core Building Blocks of CNNs

Key Layers And Their Roles



Convolution Layer:

Extracts edges, textures, and patterns using filters



Pooling Layer

Reduces size while keeping key details, adds translation invariance



Fully Connected Layer:

Performs high-level reasoning and classification

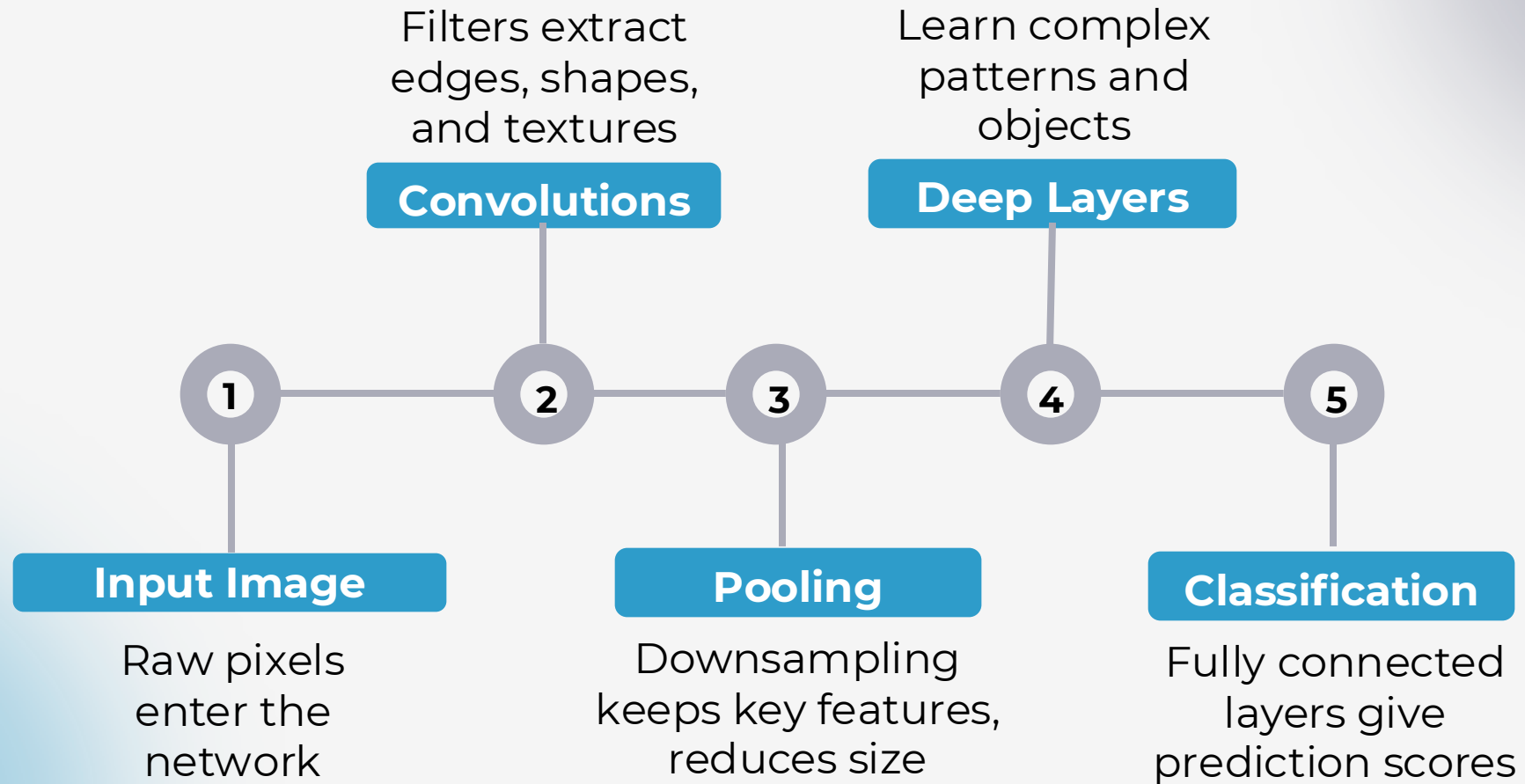


Activation Functions:

Add non-linearity (mainly ReLU) to learn complex patterns

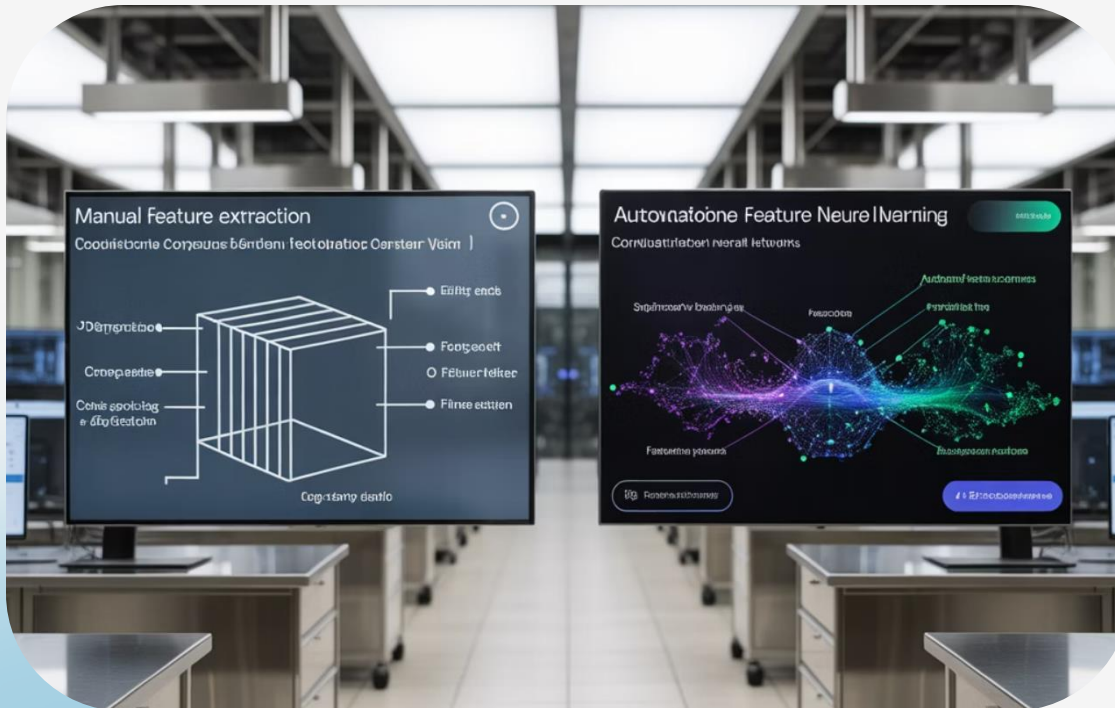
CNN Workflow: From Pixels to Predictions

Step-by-step Process



Why CNNs Dominate Computer Vision

Key Strengths



Automatic Feature Learning: No manual extraction needed

Parameter Efficiency: Shared weights reduce complexity

Spatial Understanding: Suited for images, video, and spatial data

Built-in Invariance: Robust to shifts, rotations, and scaling

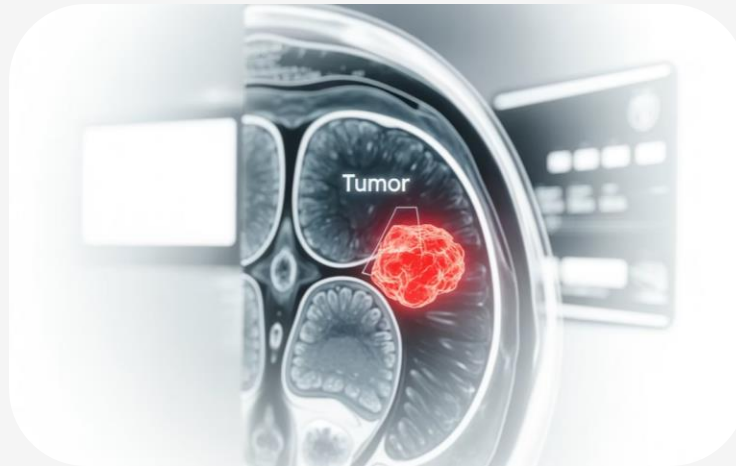
Real-World Applications of CNNs

Transforming Industries With Visual Analysis



Face Recognition

Secure logins, photo sorting, and surveillance



Medical Imaging

Detects abnormalities in X-rays, MRIs, and CT scans



Self-Driving Cars

Real-time object detection, lane tracking, and environment awareness

Key Takeaways on CNNs

How They Shape Computer Vision

CNNs don't just see images, they understand them like humans do



Biologically Inspired: Modelled on human visual processing

Hierarchical Learning: From edges to complex objects

Modern CV Backbone: Basis of ResNet, Inception, EfficientNet

Evolving Field: Driving progress in visual AI

Image Data Augmentation: Boosting Computer Vision with Artificial Diversity

The Data Dilemma in Computer Vision

Limits Of Training Data



The Problem

CNNs need large labelled datasets to prevent overfitting

The Challenge

Collecting and labelling data is costly and time-consuming

The Question

How to build strong models with limited data?

Image Data Augmentation

Expanding Datasets With Transformations



A method that artificially expand training data by applying transformations while keeping original labels

It helps models generalise to new data and reduces overfitting with limited datasets

Geometric Augmentation

Altering Images Without Changing Meaning

1 **Rotation:** Turned at random angles (0° – 360°)

2 **Flipping:** Horizontal or vertical reflections

3 **Cropping & Zooming:** Random crops and scale shifts

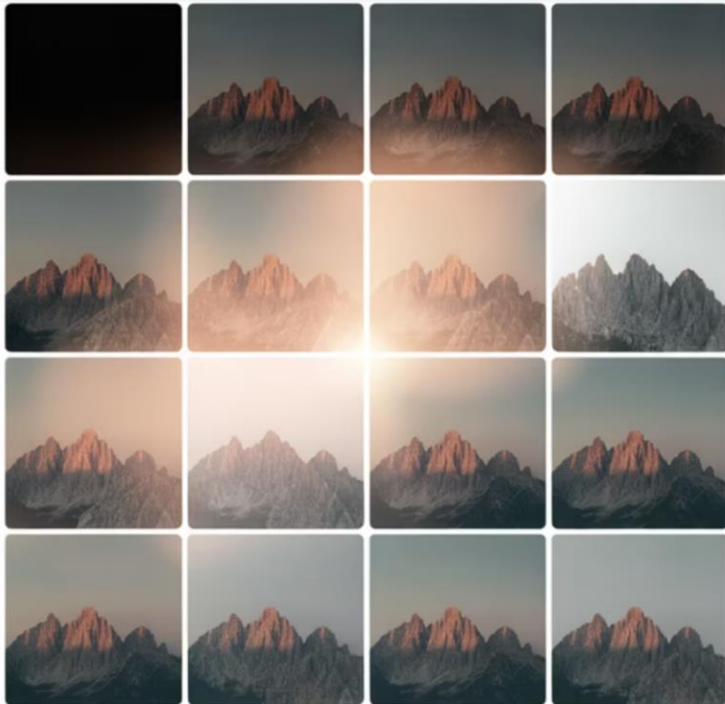
4 **Translation:** Shifted along x or y axis

Key Idea: A flipped cat is still a cat to humans and algorithms

Colour and Filter Augmentation

Simulating Lighting And Visual Changes

Simulates lighting and environment changes to make models more robust.



Color space adjustments (brightness, contrast, saturation, hue)

Apply Kernel filters like sharpening or blurring

Randomly erase parts of images

Advanced Augmentation

Creating Richer Training Samples



Image Mixing

Combines images
(Mixup, CutMix)
with shared labels



GANs

Generate new
synthetic images for
more diversity



Style Transfer

Adds artistic styles
while keeping
content intact

Medical Imaging Case Study

Augmentation In Practice

The Challenge

- Small medical datasets
- Costly, time-consuming expert labelling
- Ethical limits on data collection

The Solution: Controlled flips, zooms, colour shifts boosted accuracy by 14%



Important Caution

Some augmentations (e.g., x-axis flips) can distort medical meaning and must be avoided in specialised domains

Result: More accurate AI diagnostics using less data

Tools for Image Augmentation

Tools for Image Augmentation

TensorFlow Keras:

Preprocessing layers:

- RandomFlip
- RandomRotation
- RandomZoom
- Rescaling

Specialised Libraries:

- Albumentations
- imgaug
- torchvision

Provide flexible
augmentation APIs

Integration

Seamlessly fit into
training pipelines:

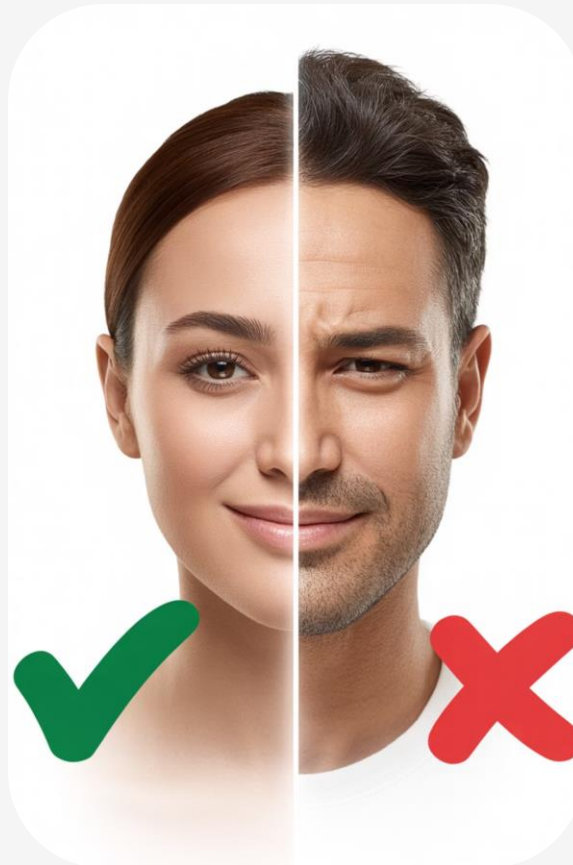
```
train_datagen =  
ImageDataGenerator(  
    rotation_range=20,  
    zoom_range=0.15,  
    horizontal_flip=True)
```

Best Practices & Challenges

Do's and Don'ts

Do's:

- Use transformations suitable for the domain
- Keep objects recognisable
- Track validation metrics to refine strategies
- Balance variety with realism



Don'ts:

- Avoid unrealistic transformations that confuse models
- Don't over-augment
- Avoid vertical flips for orientation-sensitive objects (text, faces)
- Don't assume all augmentations suit every domain

The Future of Image Data Augmentation

Emerging Approaches



AutoAugment: AI finds the best augmentation policies for each dataset



Hybrid Methods: Combine traditional augmentation with synthetic data for stronger



Democratising AI: Enable advanced vision tasks in data-scarce settings

Unlock your model's potential with augmentation today

Q&A Session



Open floor for questions and clarifications.